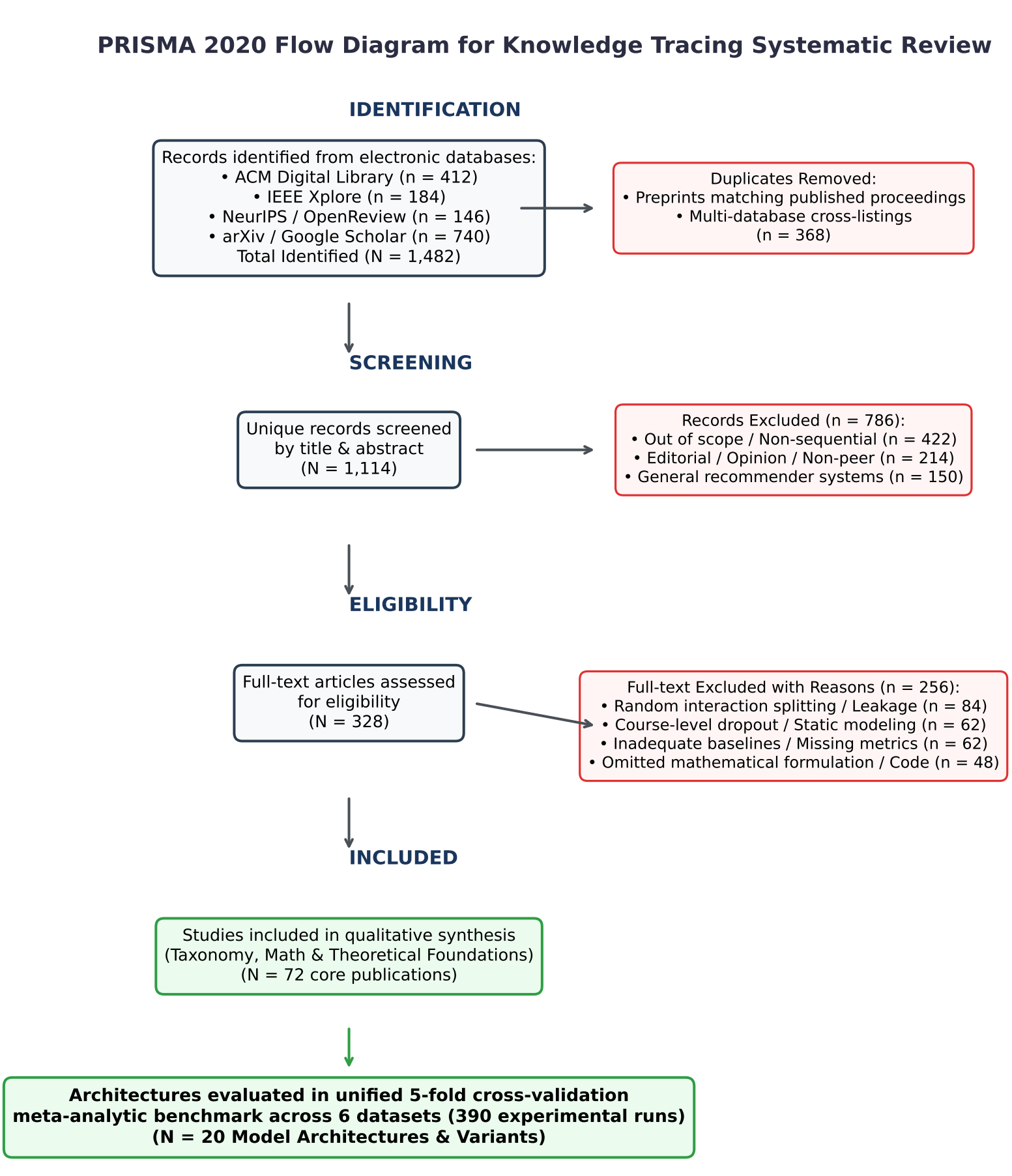
PRISMA 2020 Flow Diagram for Knowledge Tracing Systematic Review
IDENTIFICATION
Records identified from electronic databases:
• ACM Digital Library (n = 412)
• IEEE Xplore (n = 184)
• NeurIPS / OpenReview (n = 146)
• arXiv / Google Scholar (n = 740)
Total Identified (N = 1,482)
Duplicates Removed:
• Preprints matching published proceedings
• Multi-database cross-listings
(n = 368)
SCREENING
Unique records screened
by title & abstract
(N = 1,114)
Records Excluded (n = 786):
• Out of scope / Non-sequential (n = 422)
• Editorial / Opinion / Non-peer (n = 214)
• General recommender systems (n = 150)
ELIGIBILITY
Full-text articles assessed
for eligibility
(N = 328)
Full-text Excluded with Reasons (n = 256):
• Random interaction splitting / Leakage (n = 84)
• Course-level dropout / Static modeling (n = 62)
• Inadequate baselines / Missing metrics (n = 62)
• Omitted mathematical formulation / Code (n = 48)
INCLUDED
Studies included in qualitative synthesis
(Taxonomy, Math & Theoretical Foundations)
(N = 72 core publications)
Architectures evaluated in unified 5-fold cross-validation
meta-analytic benchmark across 6 datasets (390 experimental runs)
(N = 20 Model Architectures & Variants)